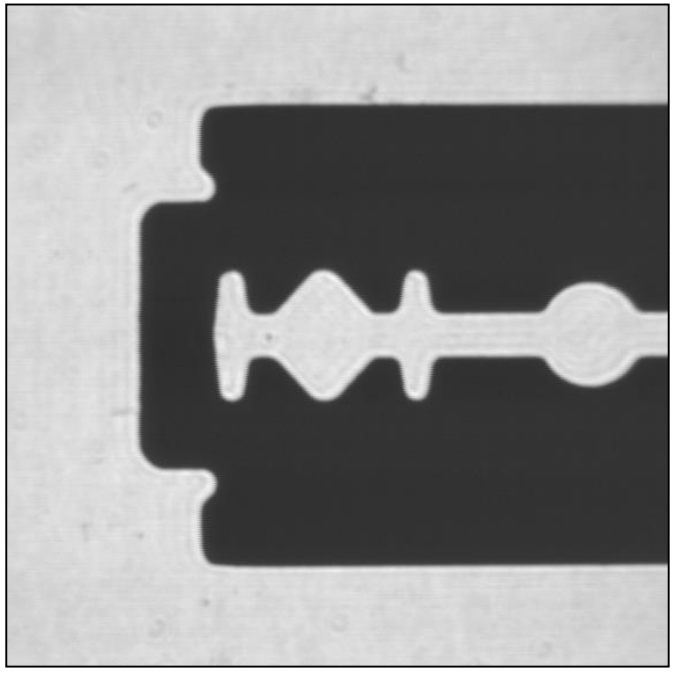
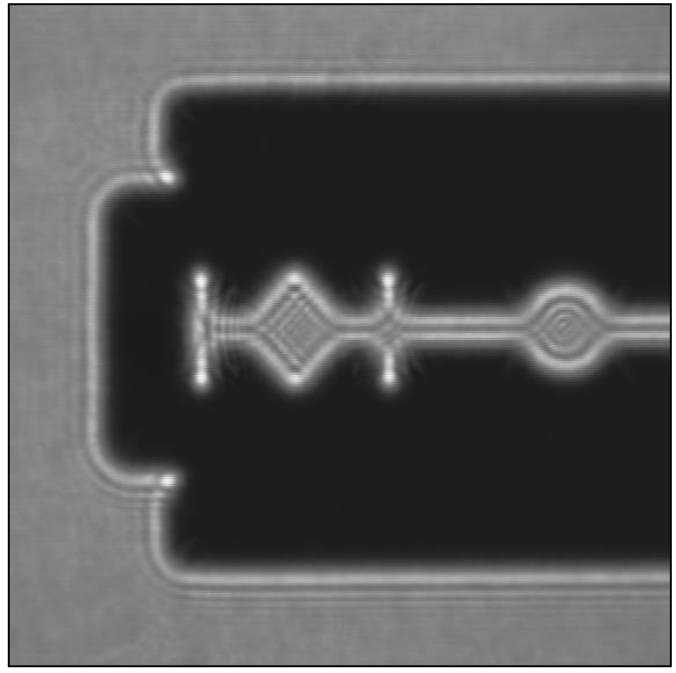
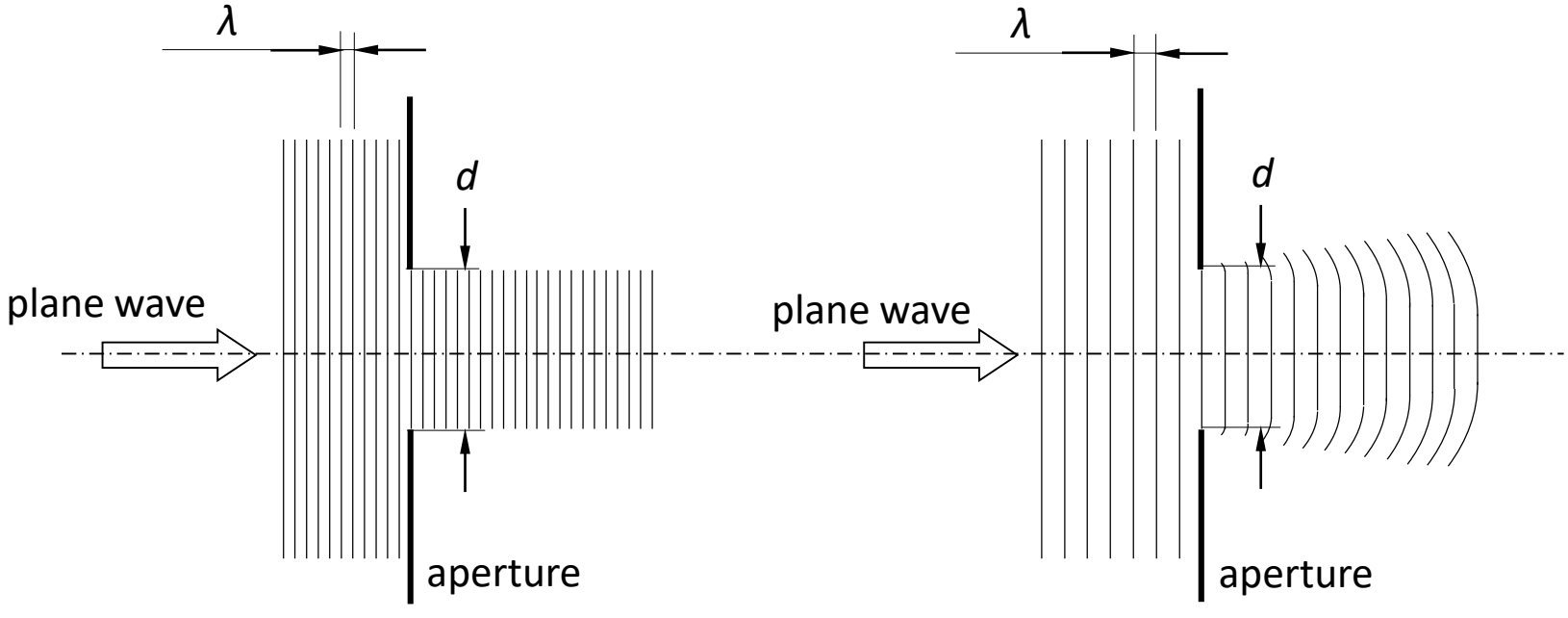
▪ Concept of diffraction ▪

diffraction: the phenomenon, when the propagation direction of light differs significantly from the one as suggested by geometrical optics (Sommerfeld).

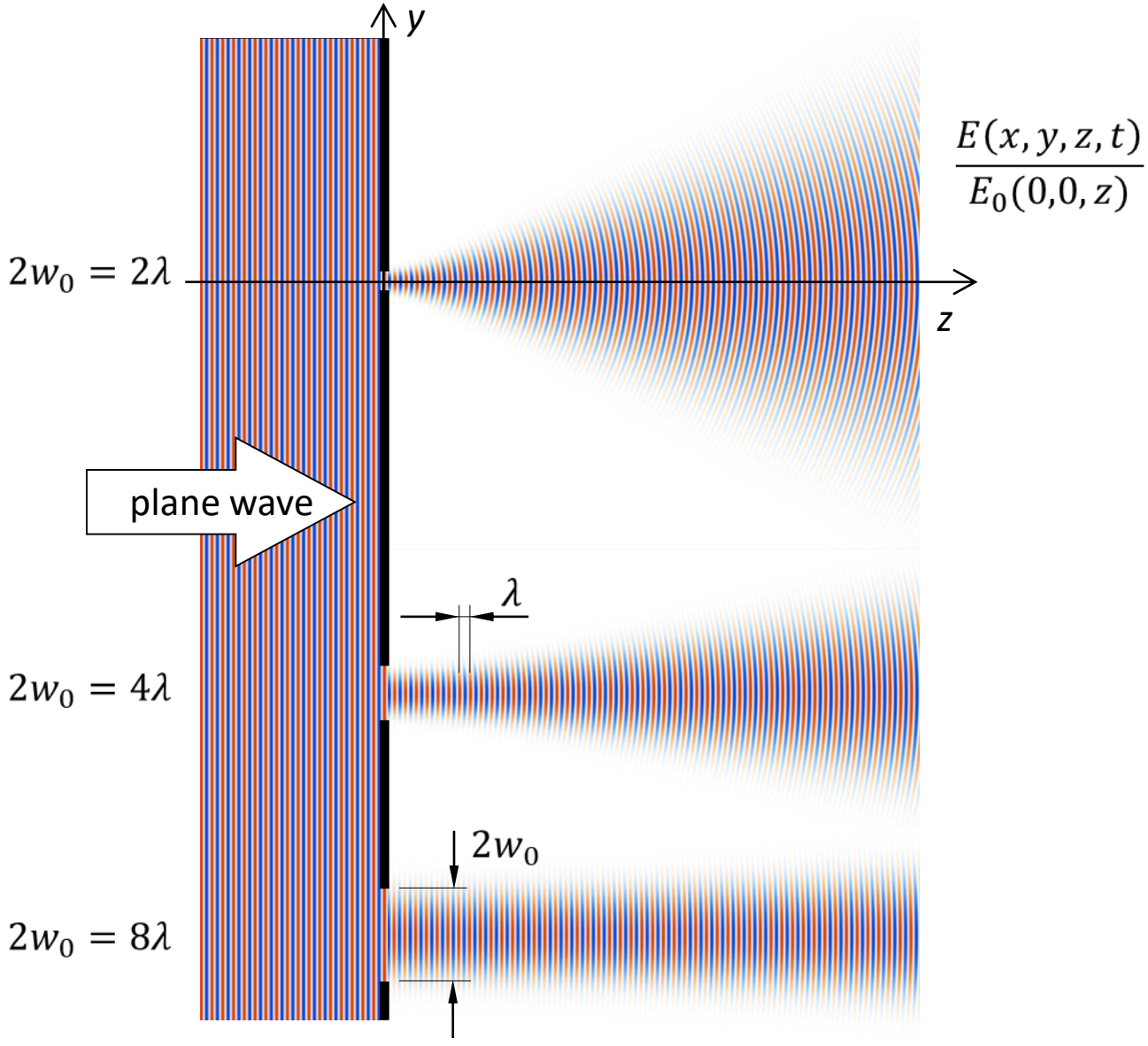




▪ Role of the wavelength ▪

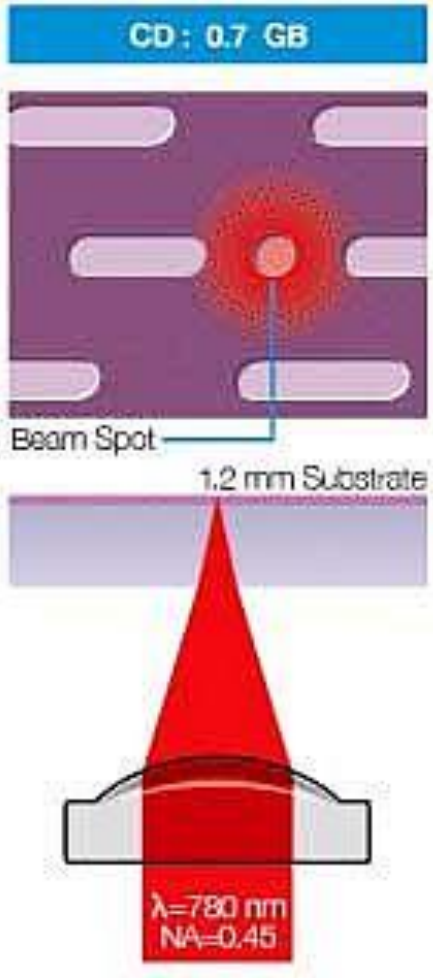


▪ Example: diffraction of a Gaussian beam ▪

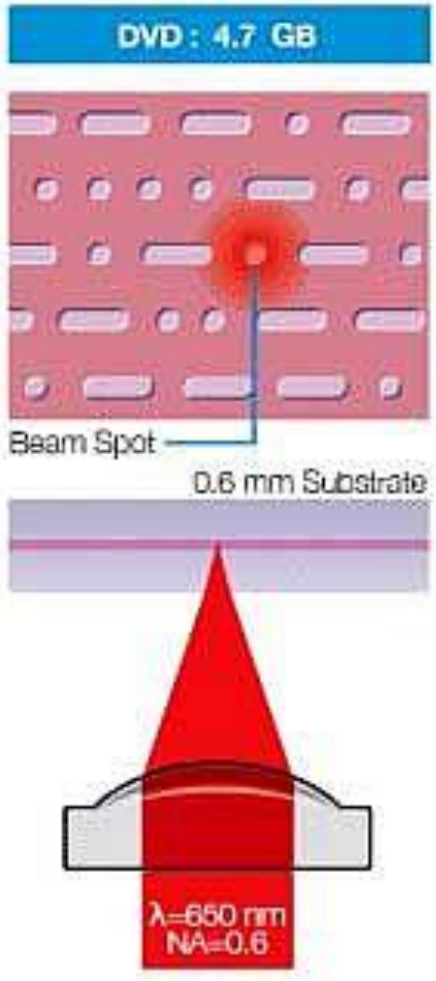


▪ Vector diffraction 1) ▪

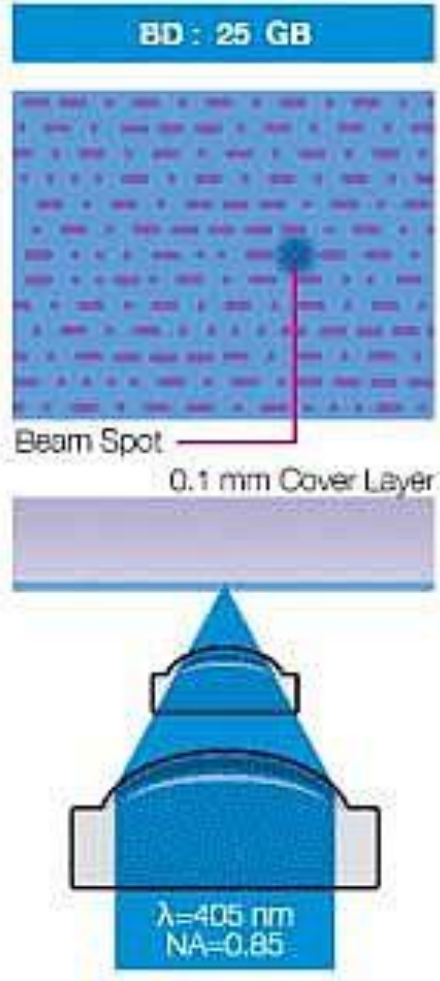
Spot size: Ø2.1 μm



Spot size: Ø1.3 μm



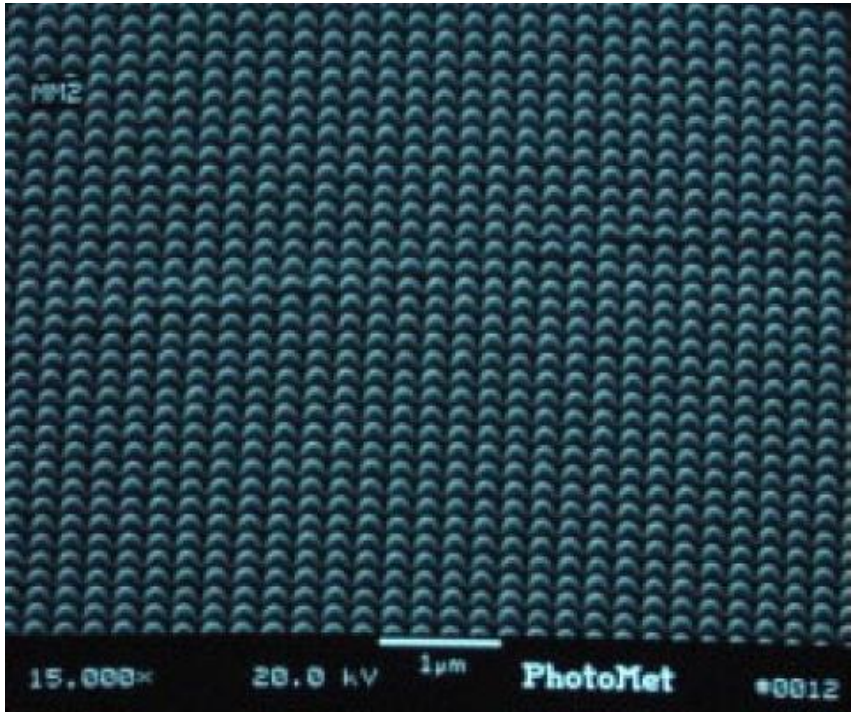
Spot size: Ø0.58 μm



$NA < 0.6$: scalar diffraction

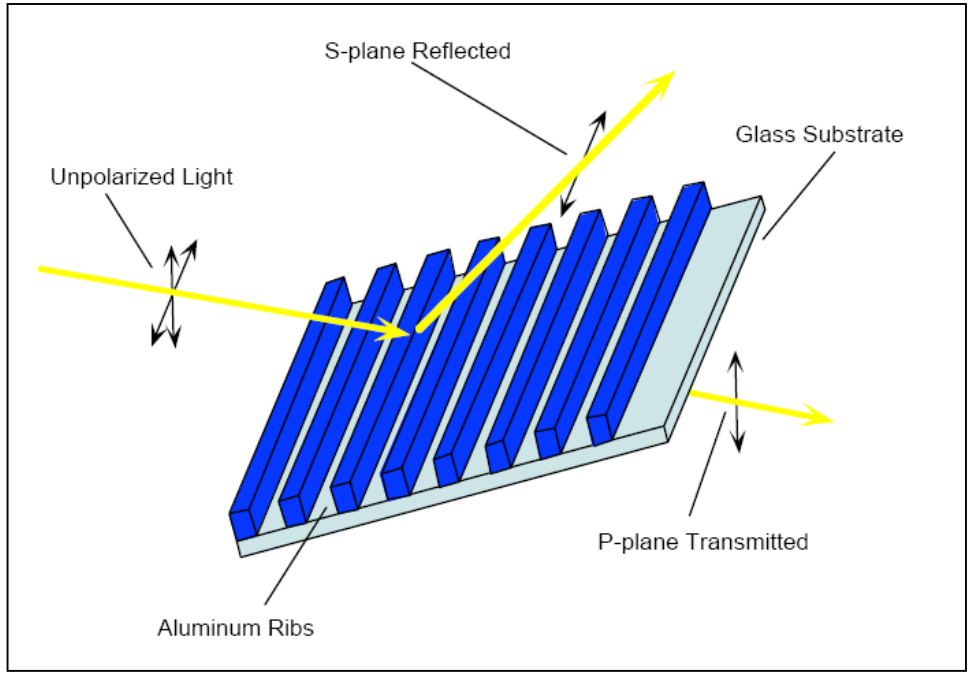
$NA > 0.6$: vector diffraction

▪ Vector diffraction 2) ▪

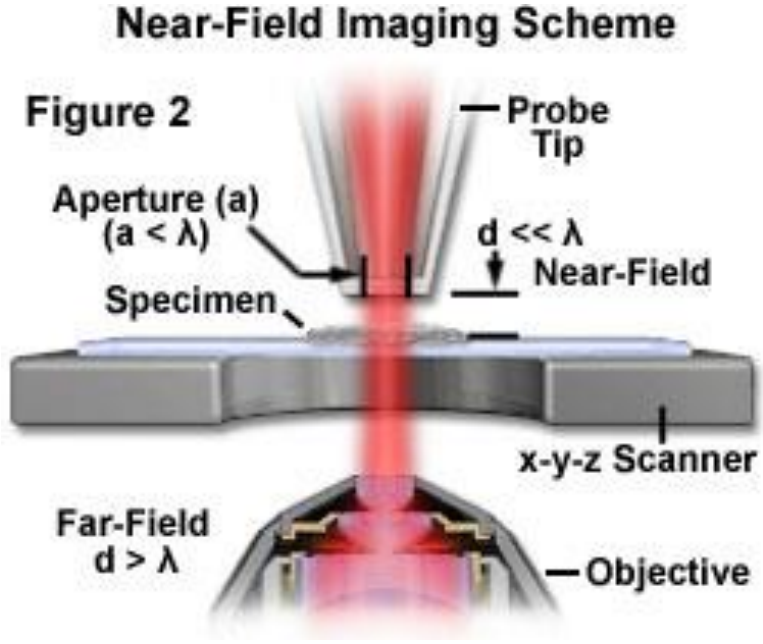


Subwavelength gratings:
“moth eye” AR coating

Subwavelength gratings: “wire grid” polarizer



▪ Vector diffraction 3) ▪



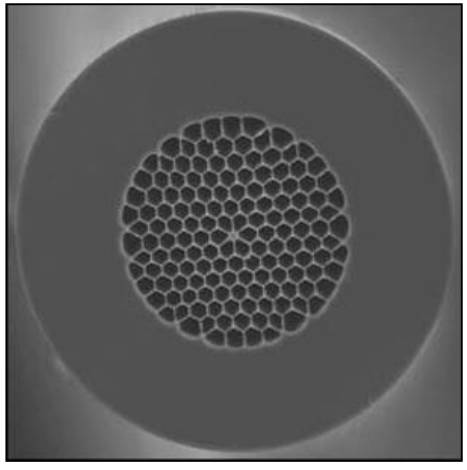
▪ Vector diffraction 4) ▪



Photonic crystals: opal

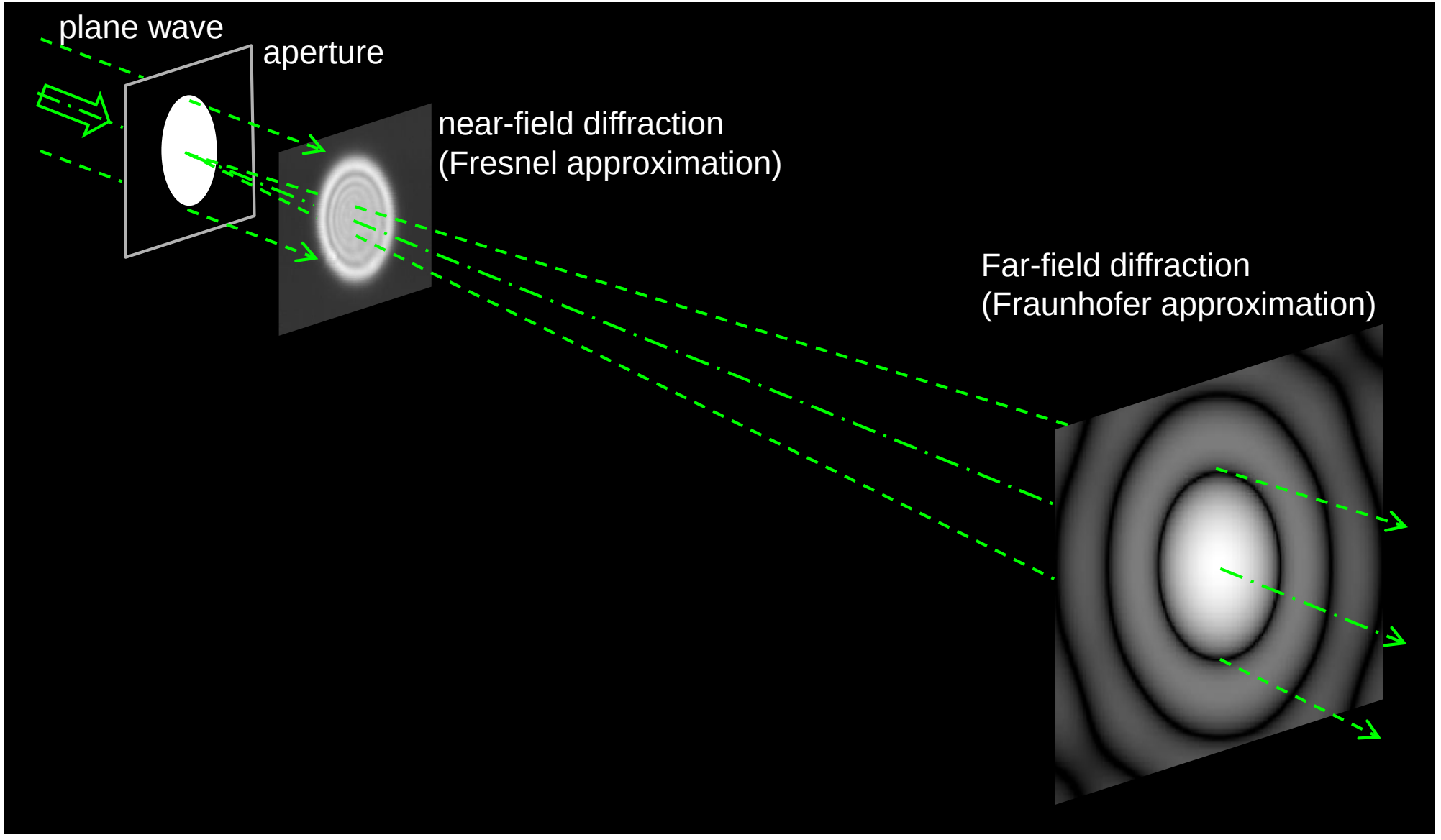


Photonic crystals: Morpho

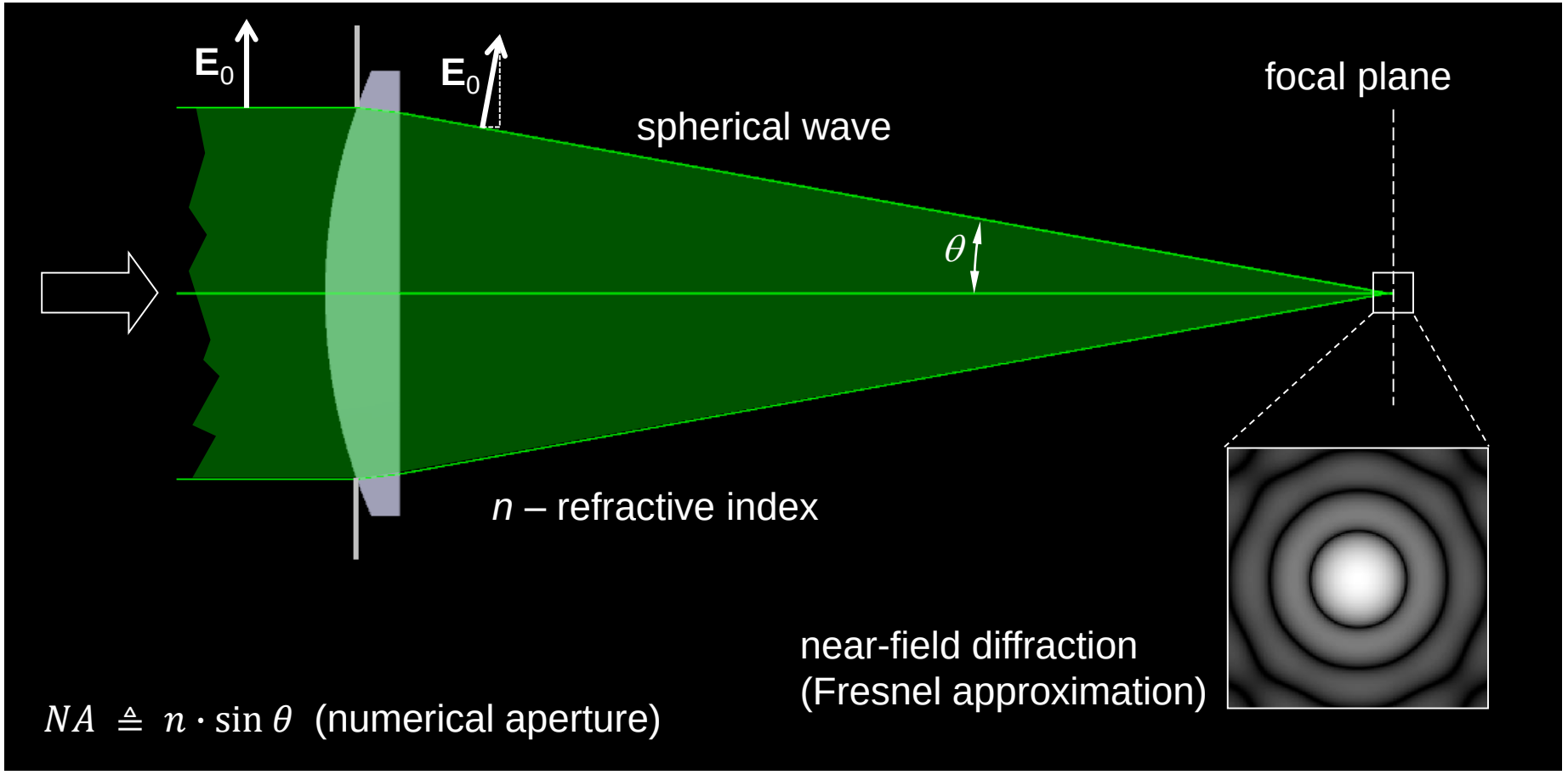


Photonic crystals: fiber optics

▪ Base problem in diffraction 1) ▪

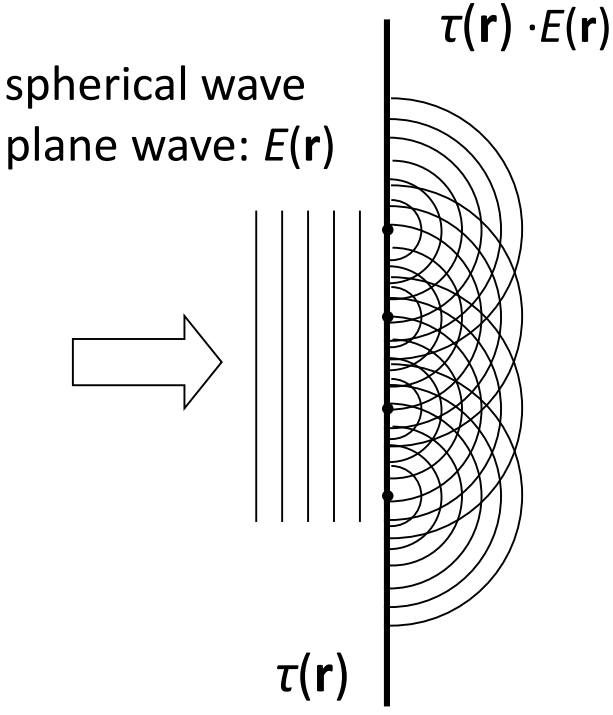


▪ Base problem in diffraction 2) ▪

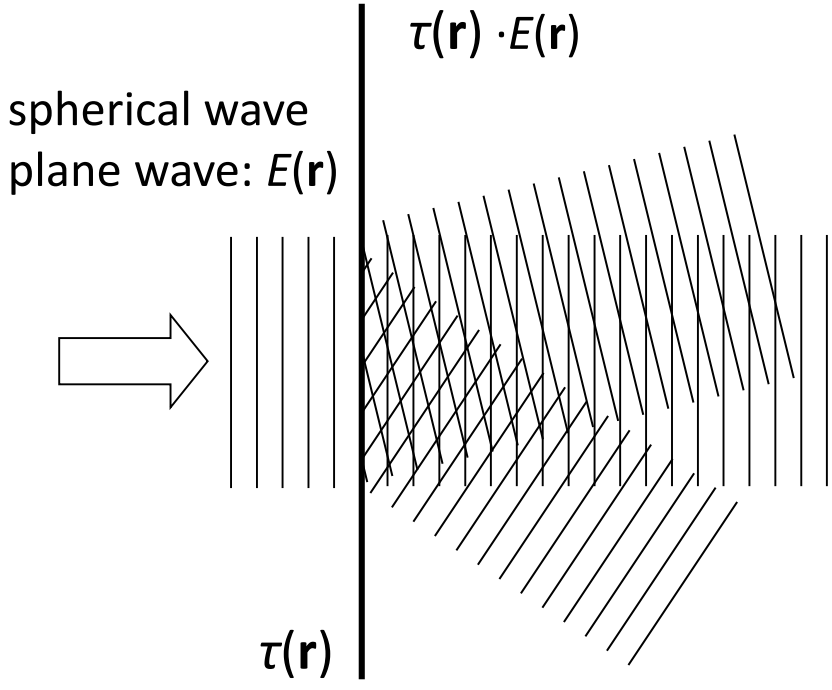


▪ Models of scalar diffraction ▪

Decomposition into point sources
(Huygens-Fresnel principle)

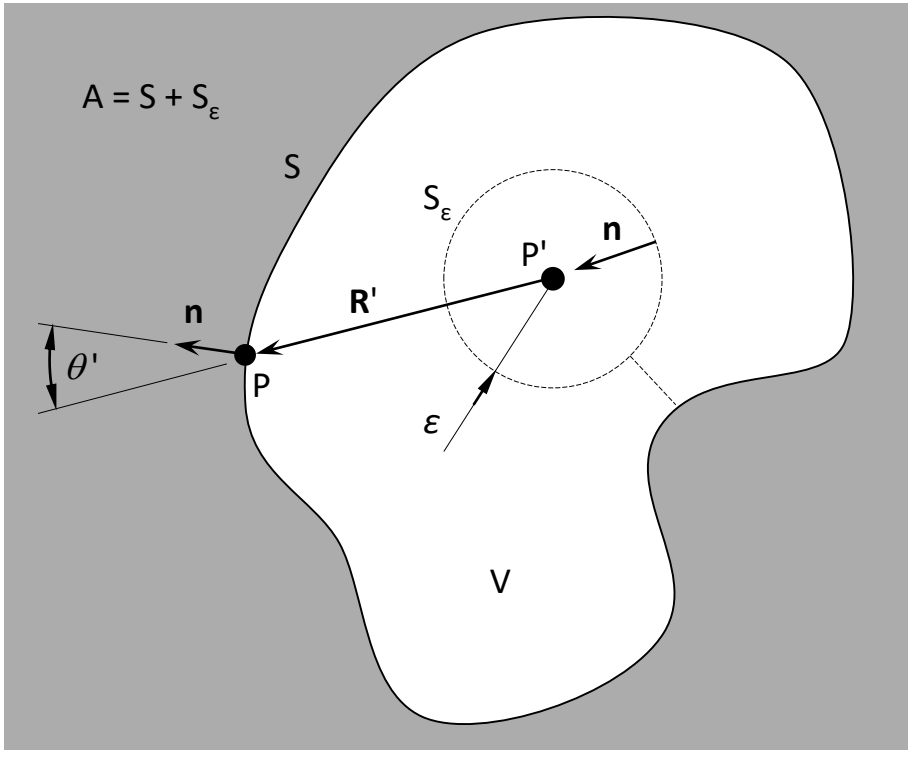


Decomposition into plane waves
(Fourier optics)

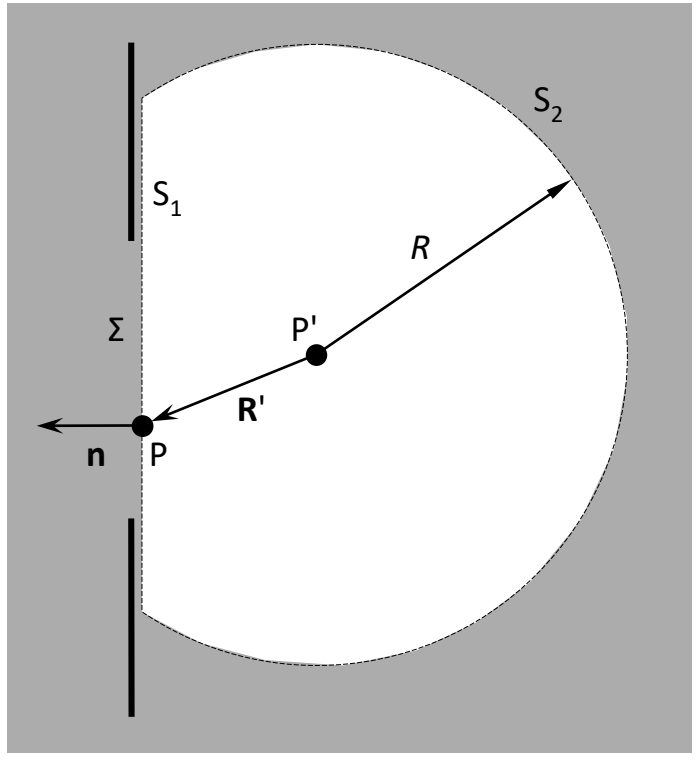


$\tau(\mathbf{r})$ – complex transmission function over the surface

▪ Definitions 1) ▪

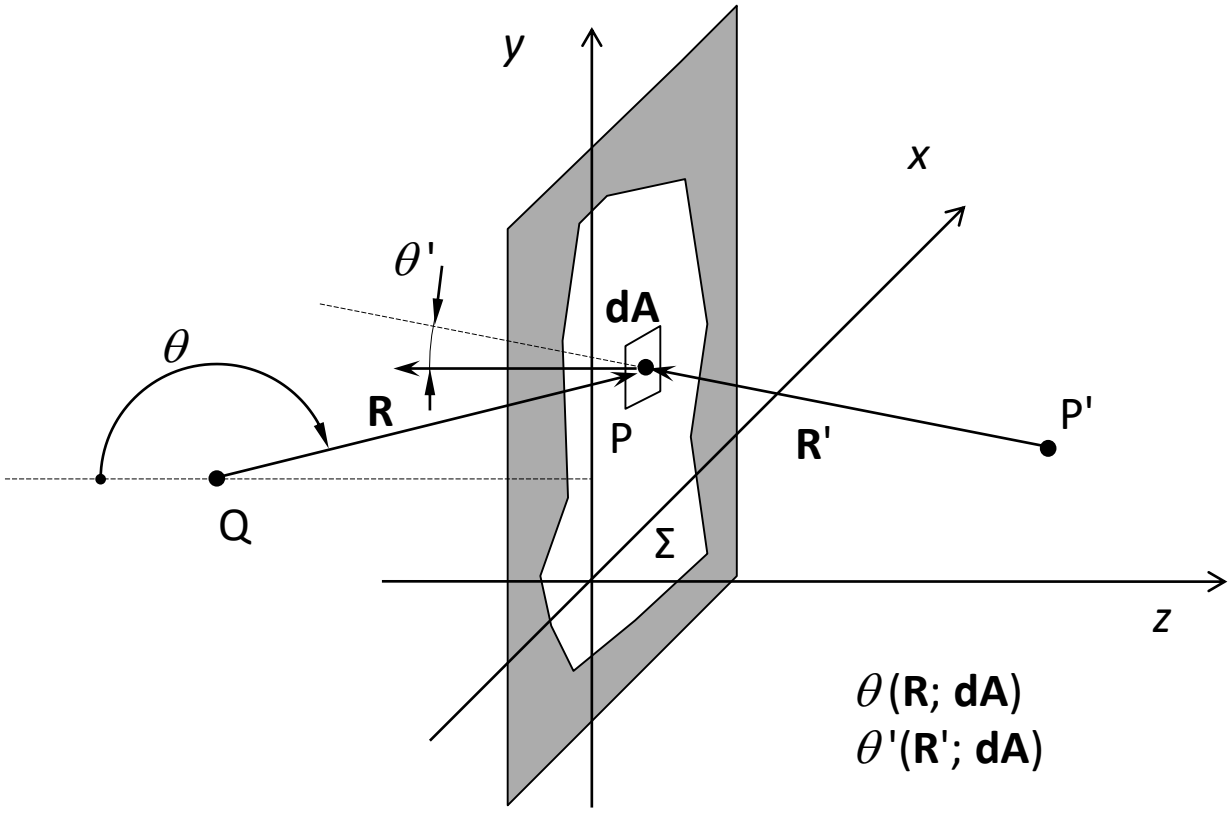


for Green's theorem

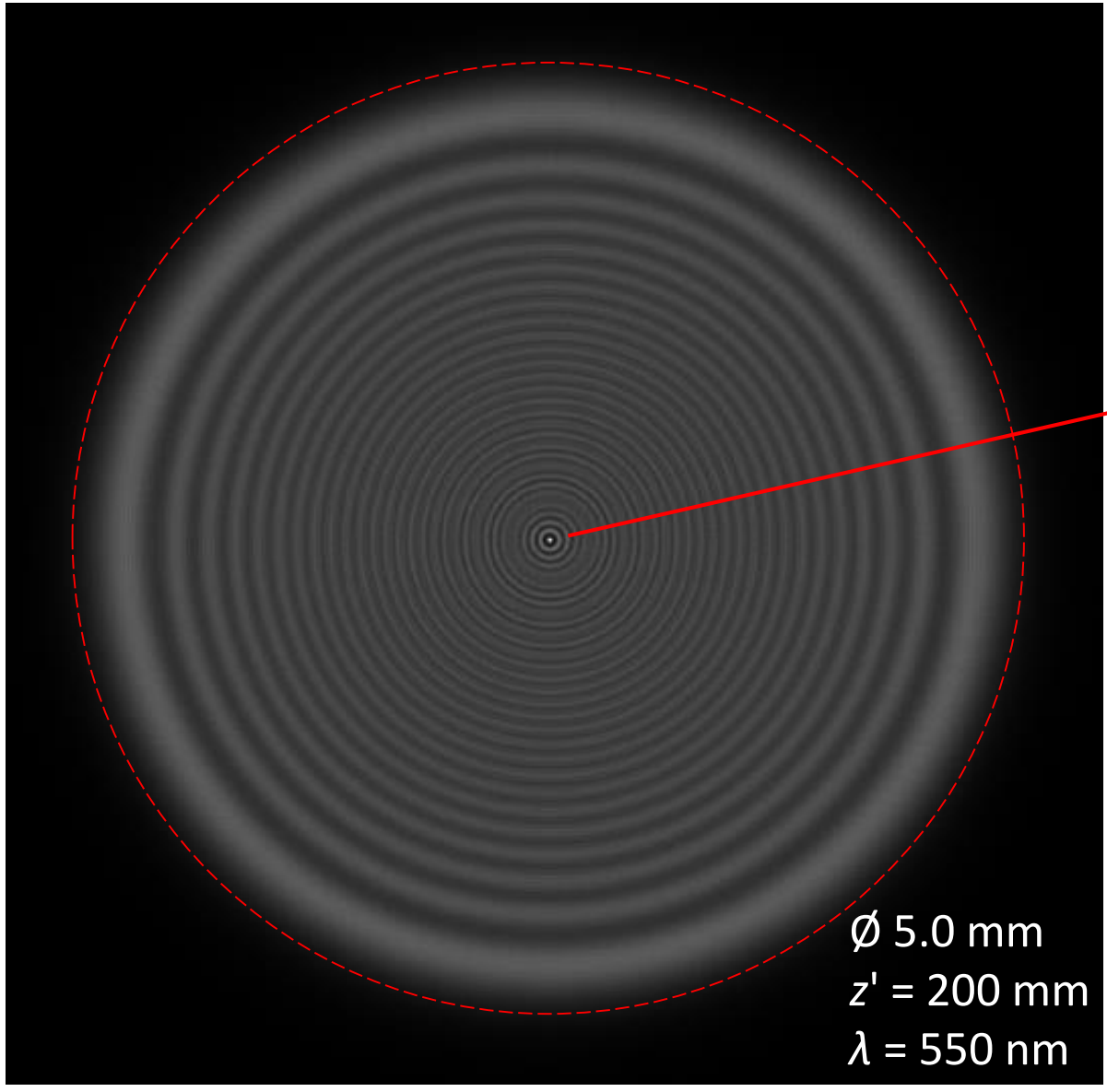


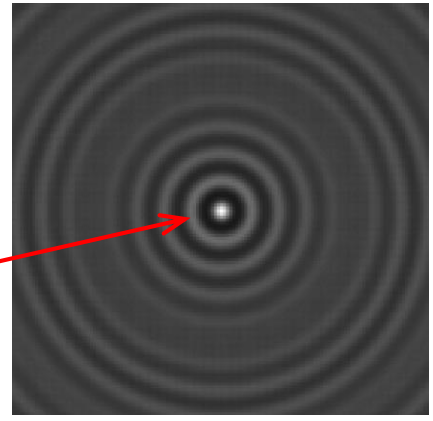
for the Fresnel-Kirchhoff diffraction integral

▪ Definitions 2) ▪



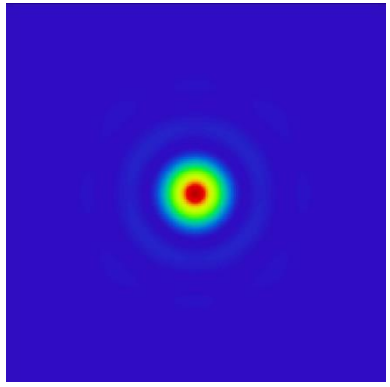
▪ Fresnel diffraction of a circular aperture ▪

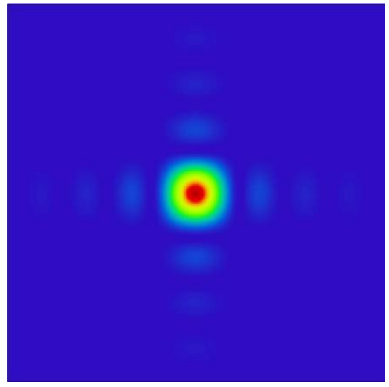


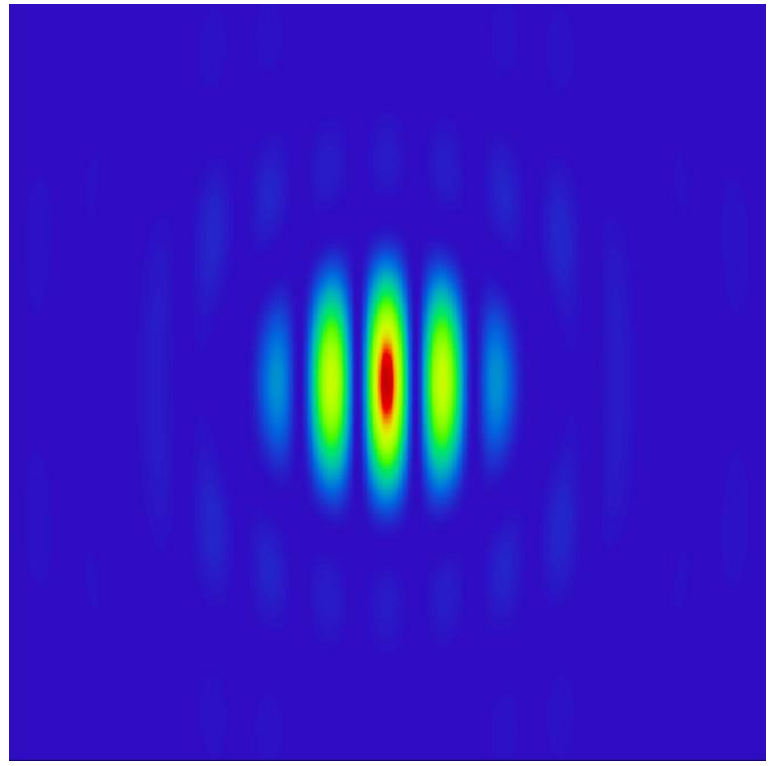


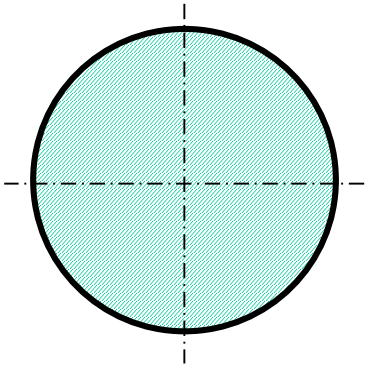
Poisson (Arago) spot

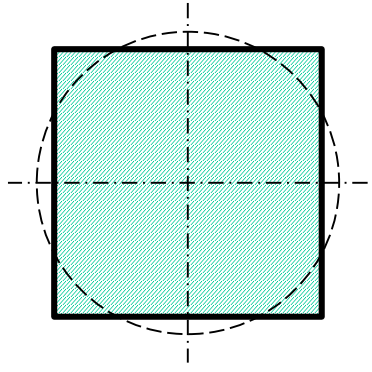
▪ Examples for Fraunhofer diffraction ▪

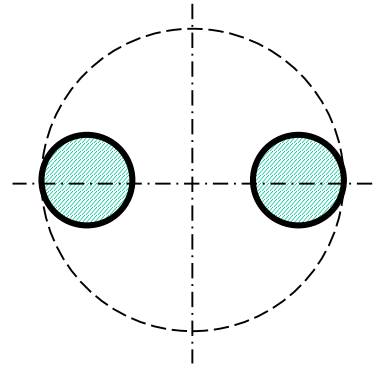




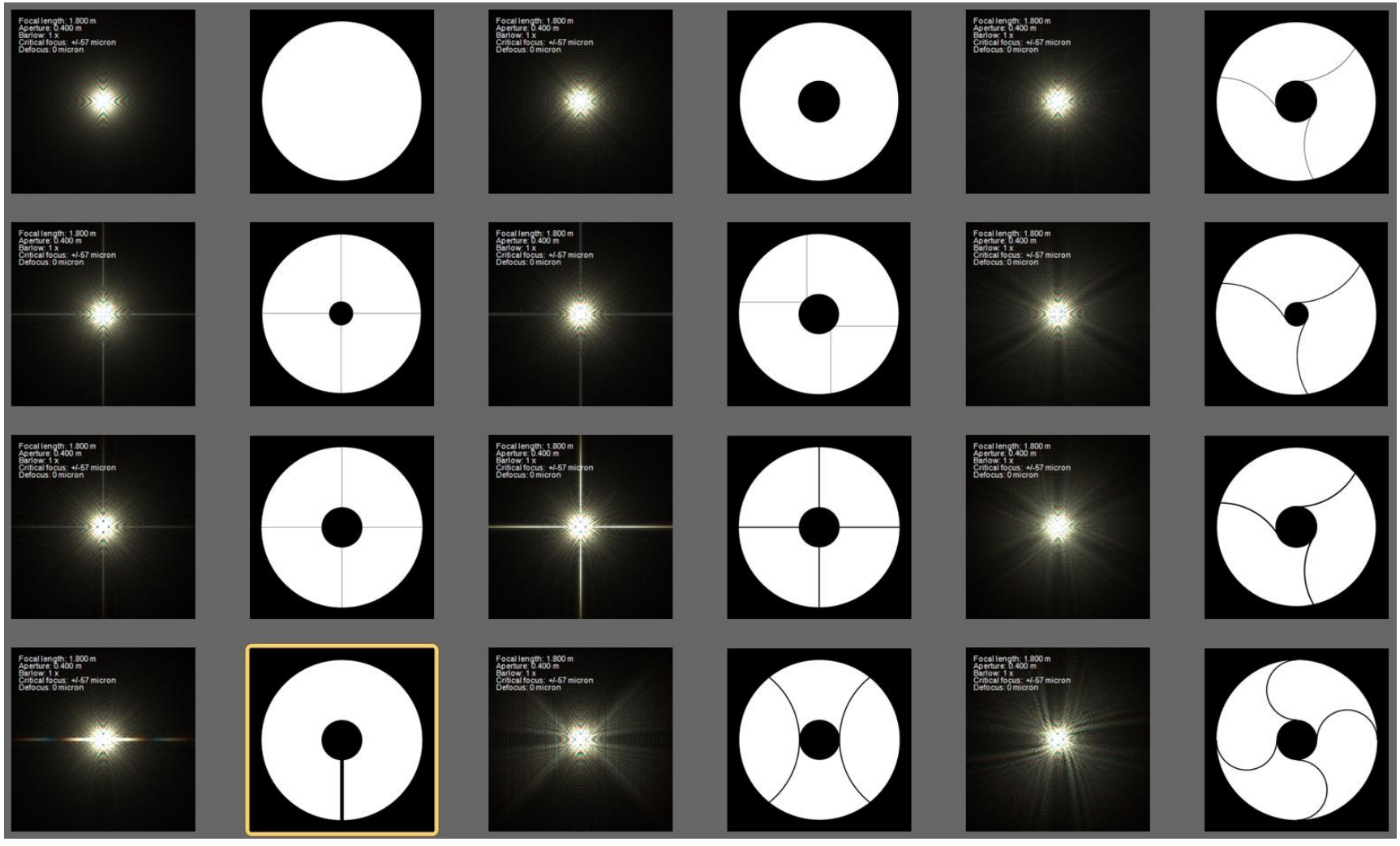






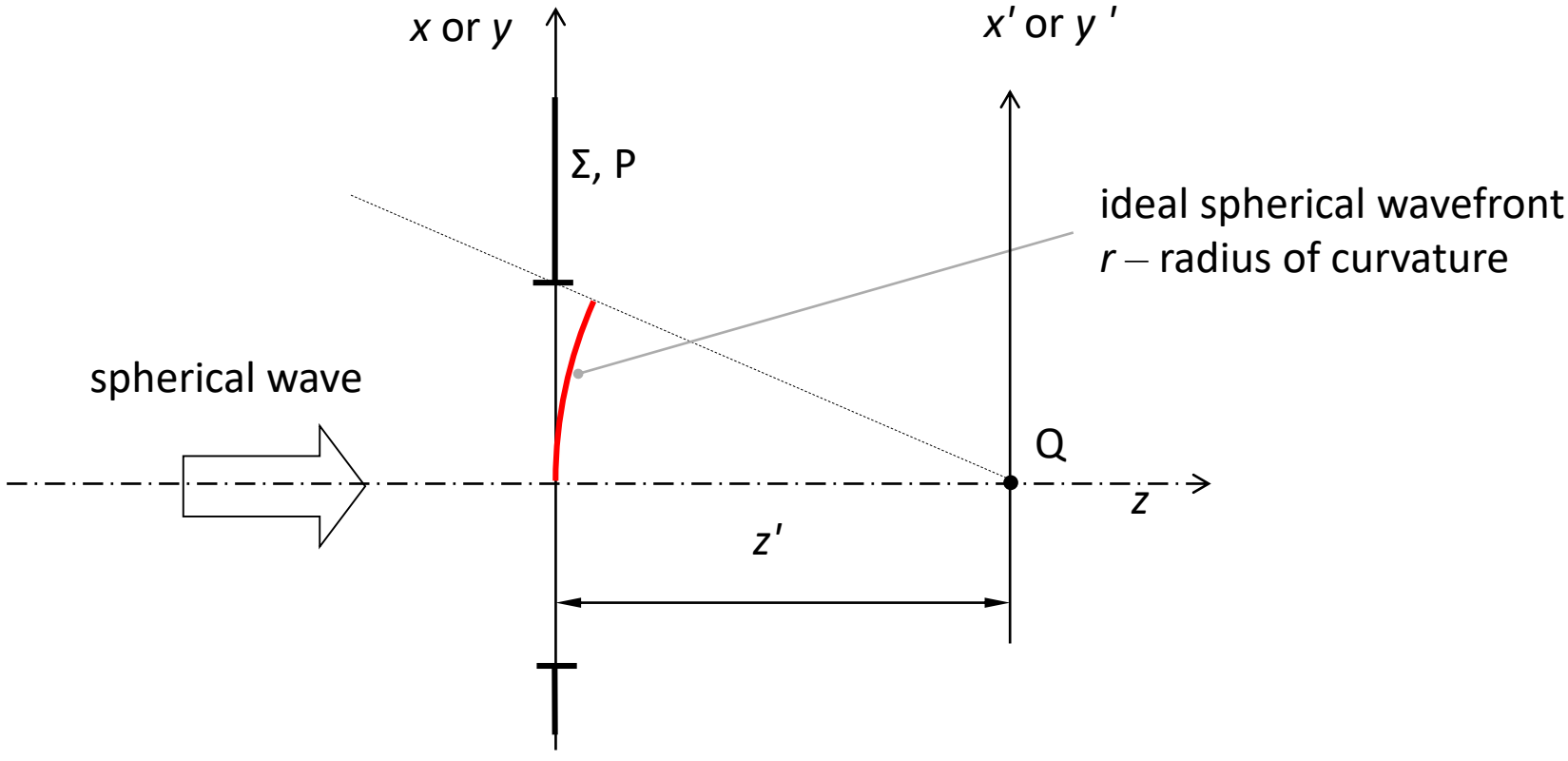


▪ Diffraction of secondary mirror spokes ▪



https://www.cloudynights.com/topic/495707-spider-and-secondary-diffraction-what-to-do-what-to-avoid/

▪ Diffraction of a spherical beam ▪



▪ Rayleigh criterion ▪

